

Nathan Maldonado

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SKILLS

Communication: Design Review, Technical Writing & Presentations, Cross-Disciplinary Collaboration

Software: Ansys (Fluent, LS-DYNA, Mechanical), MATLAB, Minitab, Python, SolidWorks, Siemens NX, VBA Macros

Technical: Additive Manufacturing (Polymers, Composites, Metals) & Post Processing, Arduino, CAD/CAM, Data Analysis, DFM, FEA, GD&T, Metrology, Milling & Turning, Nastran, Rapid Prototyping, Robotic Programming, Root Cause Analysis, PLM (Siemens Teamcenter), Statistical Analysis, Structural Analysis, Vibration Analysis

EXPERIENCE

Research Engineer, Saint-Gobain Abrasives (Norton)

August 2022 – August 2024

- Provided technical support for internal grinding and machining tests. Troubleshooted tool performance, vibration, tooling, and setup constraints with technicians, senior engineers, and manufacturing teams.
- Led three field trials at AS9100 aerospace manufacturing plants. Saved customers an average of 6% in annual manufacturing costs per operation by identifying lead-time and cost reduction strategies.
- Resolved a persistent grinding vibration issue through frequency-response testing and stiffness modeling. Guided tooling and process changes that enabled stable continuous-dress creep feed grinding.
- Partnered with an external Chemical Vapor Smoothing equipment supplier to optimize 3D-printed polymer post-processing. Provided hardware and operation guidance, project contributed \$425K in sales within six months.
- Drove abrasive product development by leading internal grinding tests. Partnered with technicians and senior engineers to provide on-floor technical support and real-time troubleshooting.
- Led five technicians in standardizing internal grinding tests and work instructions, improved repeatability and reduced internal cycle time by up to 15%. Applied GD&T and metrology to verify fixture and setup conditions.
- Managed a summer intern as a direct report, provided supervision through two grinding projects.

Research & Development Intern, Saint-Gobain Performance Plastics

May – August 2021

- Prototyped and validated a low-cost, non-intrusive flowmeter applying fluid dynamics theory and CFD modeling, enabled repeatable measurement of challenging slurry flows.
- Investigated circumferential groove designs in PTFE plain slide bearings for improved grease retention. Reduced friction and extended bearing life by 12%.

Undergraduate Research Assistant, WPI

February – December 2021

- Designed and prototyped a heated build platform for Laser-Powder Bed Fusion (L-PBF) additive manufacturing. Conducted parametric testing of build plate temperature and machine parameters to quantify effects.

EDUCATION

Worcester Polytechnic Institute (WPI), Worcester, MA

Graduation Date:

Bachelor of Science in Mechanical Engineering, GPA: 3.6/4.0

May 2022

Master of Science in Mechanical Engineering, GPA 4.0/4.0

August 2026

PROJECTS

Graduate Thesis, "FEM-SPH Simulation of Hydrodynamic Flexible Spindle Polishing," WPI

August 2024 – August 2026

- Developing a nonlinear FEM-SPH model in ANSYS LS-DYNA for polishing of additively manufactured metal internal channels. Predicting material removal and surface finish based on key input parameters.
- Validating model predictions against experimental data and refining inputs to isolate the driving mechanisms.

Senior Capstone Project, "Solar Agribots," WPI

July 2021 – May 2022

- Partnered with robotic and electrical engineering teammates to design and build a solar-powered autonomous rover for agricultural monitoring. Integrated and validated mechanical, electrical, and robotic control systems.